COVER: Knowledge-Grounded Coverage and Conflict Reasoning for Natural Language Inference in English and Portuguese

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Abstract

[TODO: Write last.]

1 Introduction

[TODO: Motivation; gap; contributions.]

2 Related Work

2.1 External knowledge in NLI

Lexical knowledge has long been used to decide whether a hypothesis word is licensed by the premise. KIM [3] feeds WordNet relations (synonymy, antonymy, hypernymy, hyponymy, co-hyponymy) as pairwise features into the co-attention, local inference and composition steps of ESIM, and Glockner et al. [12] showed that models trained on SNLI fail on pairs that differ by a single lexical substitution, precisely the cases where such knowledge should help.

With pretrained Transformers, knowledge is usually injected either as text or as entity embeddings: K-BERT [18] appends knowledge-graph triples to the input and restricts their visibility in self-attention, while KnowBert [29] and ERNIE [48] link mentions to knowledge-base entity embeddings.

Graph-based approaches retrieve a subgraph for the pair and encode it with a graph neural network whose output is fused with the sentence representation [17]. [TODO: add the references of the KGAnet and KGNLI baselines.]

A common limitation is that the retrieved evidence is used as an additional feature vector: it does not change *how* the decision is reached, and its effect on strong pretrained encoders is often small. In COVER, typed relations bias the premise–hypothesis alignment and enter the messages that each hypothesis word receives, and the resulting word-level evidence is combined by explicit logical rules.

2.2 Structure inside pretrained encoders

A second line of work integrates graphs with pretrained language models more tightly. QA-GNN [45] reasons jointly over a question–answer context node and a knowledge-graph subgraph, GreaseLM [47] exchanges information between the language model and a graph neural network at several layers, and structural adapters [32] insert graph convolutions between frozen Transformer blocks.

These methods target question answering or generation over knowledge or semantic graphs. COVER instead builds its graph over the contextual states of the encoder itself, connecting premise and hypothesis words through lexical relations, and keeps the encoder untouched: the graph operates as a separate expert whose output is added to the classifier logits.

2.3 Alignment and natural logic

Alignment between premise and hypothesis is a classic ingredient of NLI, from phrase-based aligners [22] to the soft attention of the decomposable attention model [28] and ESIM [2], whose matching features ($[\boldsymbol{x}; \boldsymbol{m}; \boldsymbol{x} \odot \boldsymbol{m}; |\boldsymbol{x} - \boldsymbol{m}|]$) we reuse in the update of Equation (6).

Natural logic [21] goes further and composes lexical relations along an alignment, taking into account the monotonicity of the context; NaturalLI [1] uses it for large-scale inference, and recent work makes the composition differentiable [8] or trains it with reinforcement learning and lexical resources [9].

These systems depend on external aligners or polarity parsers that are mostly available for English. COVER uses a learned soft alignment with a *null* target and does not model monotonicity explicitly; the weakness this causes on downward-monotone contexts is discussed in Section 4.

2.4 Atomic and probabilistic inference

The aggregation in COVER is closest to models that decide NLI from the parts of the hypothesis. Glickman et al. [11] defined the probability that a text entails a hypothesis as the product, over hypothesis words, of the probability that each word is entailed by some text word, an idea later extended by Shnarch et al. [35].

In neural NLI, SLR-NLI [37] treats hypothesis spans as logical atoms, predicts a label for each span from sentence-level supervision, and derives the pair label with rules such as "a contradiction requires at least one contradicting span"; it retains in-distribution accuracy and improves robustness when training data is scarce.

EPR [43] aligns phrases of the two sentences, predicts a label per aligned pair and induces the sentence label with differentiable fuzzy-logic formulas, and FGLR [38] applies the same principle to facts generated by a large language model.

COVER shares the decision rule of these models (any conflict yields contradiction; full support yields entailment) but differs in three respects. First, its atoms are contextual token nodes rather than spans, phrases or generated facts, so no chunker or generator is needed. Second, the alignment

and the messages of these atoms are shaped by typed lexical relations.

Third, the structured distribution is fused with, rather than substituted for, the sentence-level classifier, which keeps phenomena that the rules cannot express within reach of the encoder. [TODO: confirm in the SLR-NLI and EPR papers whether either combines its prediction with a sentence-level classifier or uses external knowledge.]

2.5 Robustness and diagnostic evaluation

NLI models exploit annotation artifacts [15] and shallow heuristics such as lexical overlap [24], so in-distribution accuracy overstates their inferential ability.

Diagnostic sets address this by annotating pairs with the phenomenon they test: the GLUE/SuperGLUE diagnostic [40, 41], which we use as AX-b, covers lexical semantics, predicate–argument structure, logic and world knowledge, and MED [44] shows that models fine-tuned on standard data handle upward but not downward monotonicity. Following this line, we report results on out-of-distribution and diagnostic test sets in addition to in-distribution accuracy, and analyse AX-b by phenomenon.

2.6 NLI for Portuguese

Portuguese NLI has been driven by shared tasks and translated resources: ASSIN [10] annotates entailment and paraphrase in news, SICK-BR [30] translates SICK, and ASSIN 2 [31] builds on SICK-BR with a binary entailment label. InferBR [39] provides three-way labels and is the main benchmark of this work.

Most recent systems fine-tune Portuguese or multilingual encoders such as BERTimbau [36], Albertina [33] and XLM-R [5], and cross-lingual evaluation typically relies on XNLI [4], which does not include Portuguese.

Lexical-semantic resources for Portuguese are comparatively rich, from Onto.PT [14] to PT-LKB [13], which merges ten resources, but to our knowledge they have not been integrated into the inference mechanism of Transformer-based NLI models. [TODO: check this claim with a targeted search before submission.] COVER uses PT-LKB in this way, and we also test whether the effect transfers between English and Portuguese.

3 Methodology

3.1 Task formulation

Given a premise $\mathcal{P}$ and a hypothesis $\mathcal{H}$, natural language inference (NLI) asks whether $\mathcal{P}$ entails $\mathcal{H}$ (E), contradicts it (C), or neither (N). Some corpora only distinguish entailment from its absence; we keep a single three-way output space $\mathcal{Y} = \{\text{E}, \text{N}, \text{C}\}$ and evaluate two-way corpora through the induced binary decision E *vs.* ¬E (Section 3.6).

Our design starts from a simple observation about the label semantics. A hypothesis is entailed when *every* part of it is supported by the premise and *no* part of it conflicts with the premise; it is contradicted as soon as *one* part conflicts; otherwise it is neutral.

This view goes back to probabilistic lexical entailment [11], where $P(\mathcal{H} \mid \mathcal{P})$ is a product over hypothesis words, and to recent atomic-inference models that predict a label per hypothesis span and combine them with logical rules [37, 43]. COVER implements it over *contextual token states* and grounds the per-word decisions in *typed lexical knowledge*.

3.2 The COVER architecture

Figure 1 gives an overview. COVER keeps the standard cross-encoder classifier and adds a structured expert whose output is a log-probability distribution over $\mathcal{Y}$. The expert has three parts: a token graph built from symbolic resources (Section 3.2.2), bipartite relational attention over contextual states (Section 3.2.3), and a coverage–conflict head with a logical aggregation (Section 3.2.4). The two are fused in logit space (Section 3.2.6).

3.2.1 Encoder

The pair is encoded jointly as `[CLS]` $\mathcal{P}$ `[SEP]` $\mathcal{H}$ `[SEP]` by a pretrained Transformer, yielding contextual states $\boldsymbol{h}_t \in \mathbb{R}^d$ for every subword t and a pooled state $\boldsymbol{h}_{[\text{CLS}]}$ (the pooler output when the encoder has one, otherwise the first token). The baseline classifier is

$$\boldsymbol{\ell}_{[\text{CLS}]} = \boldsymbol{W}_c \, \boldsymbol{h}_{[\text{CLS}]} + \boldsymbol{b}_c \in \mathbb{R}^3. \tag{1}$$

COVER is encoder-agnostic; we use RoBERTa [19] for English, BERTimbau [36] for Portuguese and XLM-R [5] for the cross-lingual setting (Section 3.6).

3.2.2 Token graph

The graph is aligned with the encoder's subword sequence, so that symbolic evidence attaches to contextual states rather than to static word embeddings. A word becomes a node if it is a content word (not punctuation and not in a language-specific stopword list) or a logical operator; the node is represented by the state of its *first* subword. We write $\mathcal{P}$ and $\mathcal{H}$ also for the sets of premise and hypothesis nodes.

Each word also receives an operator type $o_i \in \mathcal{O}$ from language-specific lexicons for English and Portuguese, with 13 types: *none*, negation, universal, existential, downward-monotone, restrictive, conditional, disjunction, conjunction, comparative, modal, temporal and number (e.g. *not, no, without* / *não, nenhum, sem* are negations).

Portuguese *se*/*caso* is tagged as conditional only at the start of a clause, since it is otherwise a reflexive pronoun or a noun.

Premise and hypothesis are connected by typed lexical relations. For every premise–hypothesis word pair, a relation $r_{ij} \in \mathcal{R}$ is assigned by the first rule that fires: (i) *number-equal* or *number-conflict* when both words are numerals; (ii) *identical* surface form; (iii) *antonym*, *hypernym*, *hyponym* or *synonym* from a lexical resource, looked up over a small set of lemma candidates of each word; (iv) *same-stem* under a light suffix stripper; otherwise *none*.

Hypernymy is directional: $r_{ij} = \textit{hypernym}$ means the hypothesis word j is more general than the premise word

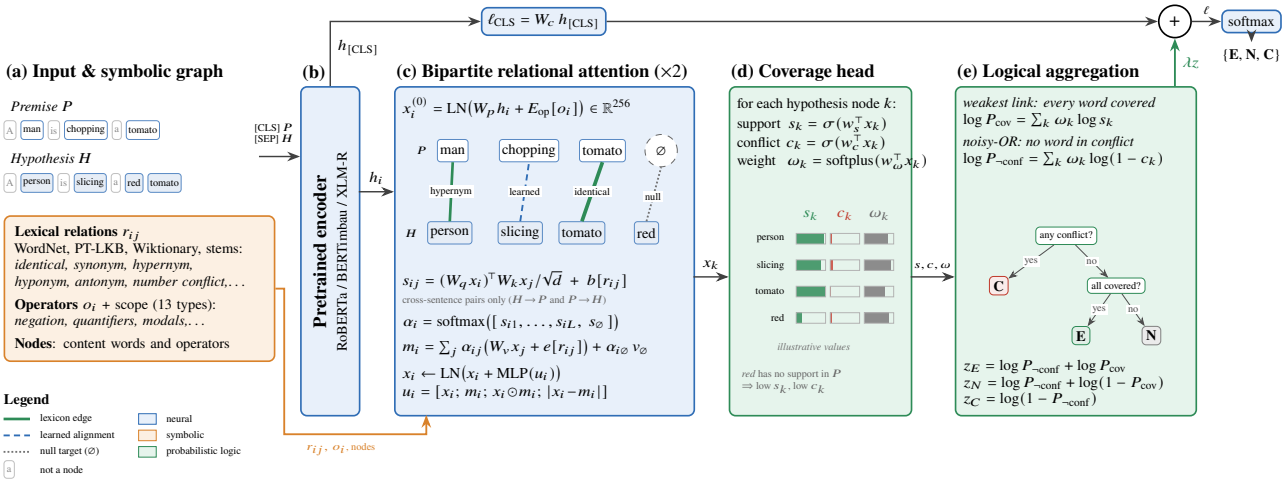


Figure 1: COVER architecture. (a) Content words and logical operators of $\mathcal{P}$ and $\mathcal{H}$ become graph nodes; typed lexical relations r_{ij} link premise and hypothesis nodes. (b) A pretrained encoder reads the pair jointly. (c) Two bipartite relational attention layers refine each node with its evidence on the other side, including a learned *null* target. (d) Every hypothesis node k receives a support s_k, a conflict c_k and an importance ω_k. (e) A weakest-link coverage and a noisy-OR conflict define a structured distribution z, added to the [CLS] logits. [TODO: update panel (a): operator scope edges are built but not used by COVER.]

i ($i \sqsubset j$), and the reverse edge r_{ji} is the inverse relation. Stopwords never receive lexical relations.

The resources are WordNet 3.0 [25] and Wiktionary [46] for English, and PT-LKB [13, 27], which merges ten Portuguese lexical resources, and Wiktionary for Portuguese. Pairs listed with both hypernym directions in the merged resources are discarded instead of guessed, since a wrong direction swaps forward and reverse entailment. From WordNet we keep single-word lemmas and the three most frequent synsets per part of speech, to limit edges from rare senses.

3.2.3 Bipartite relational attention

Node states are projected to a smaller space and typed by their operator,

$$\boldsymbol{x}_i^{(0)} = \mathrm{LN}\big(\boldsymbol{W}_p\, \boldsymbol{h}_i + \boldsymbol{E}_{\mathrm{op}}[o_i]\big) \in \mathbb{R}^{d_g}, \tag{2}$$

with $d_g = 256$. Each of $L = 2$ layers lets every premise node attend to the hypothesis nodes and every hypothesis node attend to the premise nodes (cross-sentence pairs only), plus a learned *null* target $\varnothing$ for words without a counterpart. For a node i and a node j on the other side,

$$e_{ij} = \frac{(\boldsymbol{W}_q \boldsymbol{x}_i)^\top \boldsymbol{W}_k \boldsymbol{x}_j}{\sqrt{d_g}} + b[r_{ij}], \tag{3}$$

$$\boldsymbol{\alpha}_i = \mathrm{softmax}\left([\, e_{i1}, \ldots, e_{iN},\, e_{i\varnothing}\,]\right), \tag{4}$$

$$\boldsymbol{m}_i = \sum_j \alpha_{ij}\big(\boldsymbol{W}_v \boldsymbol{x}_j + \boldsymbol{e}[r_{ij}]\big) + \alpha_{i\varnothing}\, \boldsymbol{v}_\varnothing, \tag{5}$$

$$\boldsymbol{x}_i \leftarrow \mathrm{LN}\big(\boldsymbol{x}_i + \mathrm{MLP}([\boldsymbol{x}_i; \boldsymbol{m}_i; \boldsymbol{x}_i \odot \boldsymbol{m}_i; |\boldsymbol{x}_i - \boldsymbol{m}_i|])\big). \tag{6}$$

The relation enters twice: as a scalar bias $b[r]$ on the alignment score, which steers *where* a word looks, and as a vector $\boldsymbol{e}[r]$ added to the message, which tells the receiving word *what kind* of evidence it found (e.g. an antonym rather than a synonym). The bias is initialised to 2 for identical, synonym, hypernym, hyponym, antonym and number relations, 1 for same-stem and 0 otherwise, so that lexicon edges are preferred alignments at the start of training; all relation parameters are learned.

The update in Equation (6) follows the matching features of ESIM [2]; $e_\varnothing$ and $\boldsymbol{v}_\varnothing$ are learned, and layers do not share weights. Unlike pairwise lexical features [3], the relations here act on contextual states and are refined over two rounds of alignment.

3.2.4 Coverage and conflict head

After the last layer, each hypothesis node $k \in \mathcal{H}$ is scored independently:

$$s_k = \sigma(\boldsymbol{w}_s^\top \boldsymbol{x}_k + b_s), \qquad \text{(support)} \tag{7}$$

$$c_k = \sigma(\boldsymbol{w}_c^\top \boldsymbol{x}_k + b_c), \qquad \text{(conflict)} \tag{8}$$

$$\omega_k = \mathrm{softplus}(\boldsymbol{w}_\omega^\top \boldsymbol{x}_k + b_\omega). \qquad \text{(importance)} \tag{9}$$

s_k is the probability that word k is supported by the premise, c_k the probability that it conflicts with the premise, and $\omega_k \geq 0$ how much the word matters for the decision (function-like content words can be down-weighted). Biases are initialised to $b_s = 2$ and $b_c = -3$, i.e. words start as supported and non-conflicting.

3.2.5 Logical aggregation

The word-level scores are combined by two logical rules. The hypothesis is covered if *all* of its words are supported (weakest link) and is free of conflict if *none* of its words conflicts (noisy-OR):

$$\log P_{\mathrm{cov}} = \textstyle\sum_{k \in \mathcal{H}} \omega_k \log s_k, \tag{10}$$

$$\log P_{\neg\mathrm{conf}} = \textstyle\sum_{k \in \mathcal{H}} \omega_k \log(1 - c_k). \tag{11}$$

With $\omega_k = 1$, Equation (10) is the product of word-level support probabilities of Glickman et al. [11], and Equation (11) is a noisy-OR over word-level conflicts. The three

labels follow the decision tree "any conflict ⇒ C; otherwise all covered ⇒ E; otherwise N":

$$\begin{aligned} z_{\text{E}} &= \log P_{\neg\text{conf}} + \log P_{\text{cov}}, \\ z_{\text{N}} &= \log P_{\neg\text{conf}} + \log(1 - P_{\text{cov}}), \\ z_{\text{C}} &= \log(1 - P_{\neg\text{conf}}). \end{aligned} \tag{12}$$

Since $P_{\neg\text{conf}} P_{\text{cov}} + P_{\neg\text{conf}}(1 - P_{\text{cov}}) + (1 - P_{\neg\text{conf}}) = 1$, z is a normalised log-distribution by construction; no extra softmax or threshold is needed.

Atomic-inference models apply similar rules with discrete span labels [37] or fuzzy logic over aligned phrases [43]; COVER differs in that the atoms are contextual token nodes whose alignment is shaped by typed lexical knowledge (Equations (3) and (5)), and in that the result is fused with, rather than substituted for, the sentence-level classifier. The per-word values (s_k, c_k, ω_k) also serve as an explanation of each prediction.

3.2.6 Fusion with the sentence classifier

The structured distribution is added to the [CLS] logits with a learned scale λ (initialised to 1):

$$\boldsymbol{\ell} = \boldsymbol{\ell}_{[\text{CLS}]} + \lambda\, z, \qquad \hat{y} = \arg\max_{y}\ \ell_y. \tag{13}$$

This is a product of experts: the coverage–conflict expert can overrule the encoder when its evidence is strong (e.g. a confident conflict on one word), and defers to it otherwise, so that phenomena the head cannot express are still handled by the encoder.

3.3 Training objective

We use a set-valued negative log-likelihood $\mathcal{L}_{\text{set}}(\boldsymbol{u}; Y) = -\log \sum_{y \in Y} \text{softmax}(\boldsymbol{u})_y$ over a set Y of admissible labels, applied to the fused logits and, as an auxiliary term, to the structured expert alone:

$$\mathcal{L} = \mathcal{L}_{\text{set}}(\boldsymbol{\ell}; Y) + \beta\, \mathcal{L}_{\text{set}}(z; Y), \qquad \beta = 0.15. \tag{14}$$

The auxiliary term keeps z a meaningful classifier on its own instead of a residual correction. With a single gold label, $Y = \{y^*\}$ and Equation (14) is the usual cross-entropy, which is the case in all experiments reported here.

The set formulation also allows training on corpora whose negative class merges neutral and contradiction (e.g. the NONE label of ASSIN), with $Y = \{\text{N}, \text{C}\}$; we leave this mixed-corpus training to future work.[TODO: keep or drop depending on the final experiments.]

The encoder and the new parameters are optimised with AdamW [20] and a linear warm-up/decay schedule; the graph and head parameters use a learning rate 10× larger than the encoder's, since they start from scratch.

3.4 Ablations

To identify which components produce the effect, we compare COVER with five variants that remove one component each while keeping everything else, including the auxiliary loss, identical:

w/o lexicon all relations and operators are set to *none* ($r_{ij} = \textit{none}$, $o_i = \textit{none}$): the graph and head remain, but alignment is learned from the encoder states only.

w/o graph $L = 0$: the head reads $\boldsymbol{x}^{(0)}$ directly; this also removes the relation-typed attention, hence the lexicon edges.

w/o conflict coverage only: $z_{\text{E}} = \log P_{\text{cov}}$ and $z_{\text{N}} = z_{\text{C}} = \log(1 - P_{\text{cov}}) + \log \frac{1}{2}$.

w/o coverage conflict only: $z_{\text{C}} = \log(1 - P_{\neg\text{conf}})$ and $z_{\text{E}} = z_{\text{N}} = \log P_{\neg\text{conf}} + \log \frac{1}{2}$.

MLP head the logical aggregation (Section 3.2.5) is replaced by an MLP over the mean and max of the hypothesis node states, followed by a log-softmax; this controls for the capacity of the head.

The [CLS] baseline (Equation (1) alone) is trained with the same hyperparameters.

3.5 Knowledge-graph baselines

We also compare with three knowledge-enhanced NLI architectures, re-implemented on the same encoders and trained under the same protocol and seeds. K-BERT-style injection [18] appends the retrieved relations to the premise as text, with a 160-token window so that they are not truncated. [TODO: add KGAnet and KGNLI references.]

The KGAnet-style model encodes a relation-labelled word graph with a relation-aware graph attention network, and the KGNLI-style model keeps separate semantic and knowledge branches. All three take their knowledge from rule-based relations (numbers, operators, scope) and English WordNet, so on Portuguese text their lexical evidence reduces to word identity.

3.6 Experimental protocol

3.6.1 Settings and data

We evaluate three settings (Table 1). *English*: RoBERTa-large [19] trained on SICK [23], tested on the SICK test set, on the SuperGLUE diagnostic AX-b [41], which annotates pairs with linguistic phenomena (lexical semantics, predicate–argument structure, logic, knowledge), and on five challenge sets described below.

Portuguese: BERTimbau-base [36] trained on InferBR [39] [TODO: check InferBR citation], tested on the InferBR test set and, out of distribution, on the ASSIN [10], ASSIN 2 [31] and SICK-BR [30] test sets.

Cross-lingual: XLM-R-base [5] trained on SICK or on InferBR and tested on all eleven test sets, with the union of the English and Portuguese lexicons in both directions. Relation types are shared across languages, so relation parameters learned in one language are applied to edges of the other.

The English challenge sets each isolate one kind of inference. Breaking NLI [12] changes a single word of an SNLI premise (antonyms, co-hyponyms, hypernyms), a direct test of lexical knowledge and of the conflict detector. HANS [24] targets the lexical-overlap heuristic that a coverage-based decision could learn: every hypothesis

Dataset	Lang.	Train	Dev	Test	Labels
SICK	EN	4,439	495	4,906	3-way
AX-b	EN	–	–	1,104	2-way
Breaking NLI	EN	–	–	8,193	3-way
HANS	EN	–	–	30,000	2-way
ConjNLI	EN	–	–	623	3-way
MED	EN	–	–	5,382	2-way
FraCaS	EN	–	–	335	3-way
InferBR	PT	8,190	633	1,705	3-way
ASSIN	PT	–	–	4,000	2-way[†]
ASSIN 2	PT	–	–	2,448[‡]	2-way
SICK-BR	PT	–	–	4,906	3-way

Table 1: Datasets. Only SICK and InferBR are used for training; all other sets are used for testing only. [†]Paraphrase pairs are counted as entailment. [‡]1,578 pairs for models trained on SICK (pairs translated from SICK training or trial data removed).

word occurs in the premise, but half of the pairs are not entailments.

ConjNLI [34] tests conjunctive sentences; we use its labelled development set, since the test labels are not public. MED [44] tests upward and downward monotonicity, which COVER does not model explicitly. The FraCaS suite [6] covers formal semantic phenomena such as quantifiers, plurals and attitudes; multiple premises are concatenated, and the 11 problems without a definite answer are discarded.

SICK-BR is the Portuguese translation of SICK; its test partition is aligned pair by pair with the SICK test set, so the same pairs can be compared in both languages. ASSIN 2 was built from SICK-BR, and about 36% of its test pairs (870 of 2,448) are translations of SICK training or trial pairs. Models trained on SICK are therefore scored on the remaining 1,578 pairs of ASSIN 2.

3.6.2 Training details

Hyperparameters are fixed in advance and shared by all variants of a setting; no per-variant tuning is performed, so that differences reflect architecture rather than search budget. We train for at most 8 epochs with batch size 32, weight decay 3.5×10^{-4}, dropout 0.103, gradient clipping at 1.0, and select the checkpoint with the best development macro-F1, with early stopping after 3 epochs without improvement.

The learning rate is 10^{-5} with 6% warm-up for RoBERTa-large, which is unstable at larger rates on small datasets [26], 2.04×10^{-5} with 2.75% warm-up for BERTimbau, and 2×10^{-5} with 6% warm-up for XLM-R. [TODO: max lengths; gradient accumulation for RoBERTa-large (2×16, same effective batch).]

3.6.3 Evaluation

For three-way test sets we report macro-F1. Because AX-b, HANS, MED, ASSIN and ASSIN 2 only distinguish entailment from non-entailment, we also report, for every test set, the Matthews correlation coefficient (MCC) of the binary decision E *vs.* ¬E obtained by merging N and C predictions. MCC is the official AX-b metric and is robust to the label imbalance of these sets.

We further break results down by the annotations of each set: phenomenon in AX-b, substitution category in Breaking NLI, heuristic and subcase in HANS, monotonicity direction in MED and section in FraCaS. With few pairs in some groups (18–140 per AX-b phenomenon), these breakdowns are exploratory.

Every system is trained with the same 10 seeds (1001–1010), and results are paired by seed. The seed fixes all random number generators at the start of a run; because variants create different numbers of parameters, initialisation and data order are not identical across variants for the same seed, so pairing blocks run-level variance only partially.

We compare two systems with the two-sided exact Wilcoxon signed-rank test on the 10 per-seed differences [42] and report the mean paired difference with a 95% percentile bootstrap interval over seeds. Within the ablation family, p-values are corrected with Holm's method [16]. Rankings of more than two systems use the Friedman test with the Nemenyi post-hoc test [7].

4 Results

[TODO: Main comparison; ablation (Table **??**); AX-b phenomena; cross-lingual transfer.]

5 Conclusion

[TODO:]

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